Homework #1 — Probability and Counting (25 pts)

MATH 241 — Probability, Fall 2026

Due Date: Friday, 9/11 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as a .pdf format. Any submissions not in this format are subjected to point deductions.
- Responses can be hand-written or typed.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- Refer to syllabus for late work deductions.

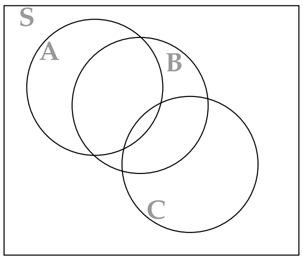
Background

The first homework will give you practice on using set notation, solving probabilities in a naive fashion, and identify/calculate probabilities of various subsets of events in a sample space.

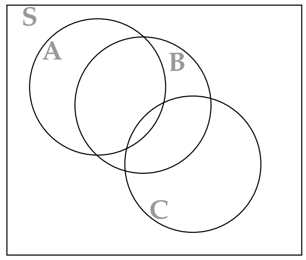
Part I: Venn Diagram (4 pts)

For each set operation below, you will shade in the corresponding area(s) of the Venn diagram.

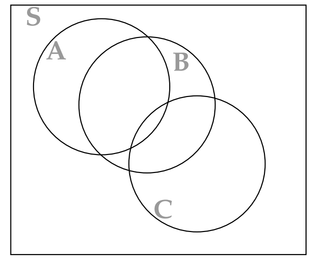
(a) $A^C \cap (B \cup C)$



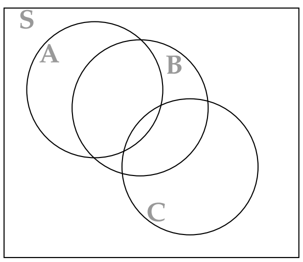
(b) $B \backslash (A \cap C)$



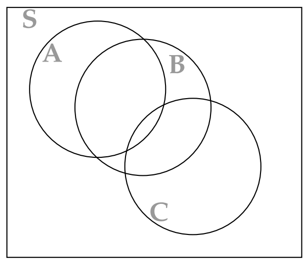
(c) $\emptyset^C$



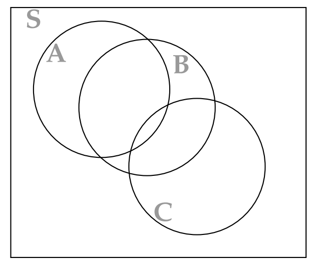
(d) $(A \cup B \cup C)^C$



(e) $(A \cap B) \cup (B \cap C) \cup (C \cap A)$



(f) $(A^C \cap B) \cup (C^C \cap A)$



Part II: 3 Rolled Dice (4 pts)

Suppose I roll 3 dice at the same time, where X is the outcome of the first die, Y is the outcome of second die, and Z is the outcome of the third die.

(a) What is the sample space of this experiment? Consider the limitations of the dice when writing out the sets.

(b) What is the cardinality of the sample space of this experiment?

(c) Describe the following events in words and in terms of the outcomes (elements) of the sample space corresponding to the three rolls. (Ex: Let (x, y) be the outcome that the first die rolls $X = x$ and the second die rolls $Y = y$. Then the event that the two rolls sum to four is $\{X + Y = 4\} = \{(1, 3), (2, 2), (3, 1)\}$.

- (i) $\{X + Y + Z = 4\}$
- (ii) $\{X > Y + 3\}$
- (iii) $\{X = Y - 1 = Z\}$

(d) Find the probability of each of the aforementioned events occurring:

- (i) $\{X + Y + Z = 4\}$
- (ii) $\{X > Y + 4\}$
- (iii) $\{X = Y - 1 = Z\}$

Part III: Football (4 pts)

Congratulations, you have just been hired as the head coach of the New York Jets. You decide to keep the play-calling very simple on the offense: all running plays.

Your running back has an equal chance of either gaining -2, 1, or 4 yards in a single play. You have 4 attempts to get 10 yards.

(a) Identify the number of possible outcomes within four plays.

(b) Identify all four-play outcomes that allow you to gain 10 yards. In other words, which permutations of four-play outcomes sums up to 10 or more?

(c) What is the probability of getting at least 10 yards within 4 plays?

Part IV: Club (4 pts)

A census has been done on all 137 members of an arts & crafts club in the Hudson Valley. We have the following characteristics of these members:

- 58 of them are above the age of 60
- 100 of them enjoy drawing
- 47 of them are above the age of 60 and enjoy drawing

Find the following probabilities of...

(a) being below the age of 60 in this club

(b) being above the age of 60 and not enjoying drawing in this club

(c) being below the age of 60 and enjoying drawing in this club

(d) being above the age of 60 or enjoying drawing in this club

Part V: Textbook Problems (9 pts)

From *Introduction to Probability, Second Edition*:

- Chapter 1: Q34, Q55